CASE STUDY

Demonstration of Network Communication Models

(OSI / TCP-IP)
Using Cisco Packet Tracer

A practical case study on packet communication between two computers

Introduction

Computer networks allow devices to exchange information using a set of communication rules and protocols. This case study demonstrates how data travels from one computer to another through the OSI and TCP/IP communication models. Cisco Packet Tracer is used to create a small network consisting of two PCs and a switch. The packet flow is then observed in Simulation Mode.

Case Study Scenario

Consider a small computer laboratory in which two computers need to communicate with each other through a network switch. PC0 acts as the source device and PC1 acts as the destination device. The objective is to verify connectivity and observe how a packet passes through different network layers.

Aim

To understand and demonstrate how data travels through the OSI model and TCP/IP model using Cisco Packet Tracer.

Objectives

- Create a simple network using two PCs and a 2960 switch.
- Configure IP addresses for the two computers.
- Verify communication using the ping command.
- Observe packet movement using Cisco Packet Tracer Simulation Mode.
- Understand the functions of the seven OSI layers.
- Relate the OSI layers to the four-layer TCP/IP model.
- Observe IP, MAC address, and protocol information inside a packet.

Software and Components Required

- Cisco Packet Tracer
- Two PCs (PC0 and PC1)
- One Cisco 2960 switch

- Copper straight-through cables

Network Configuration

Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.20	255.255.255.0

Default gateway is not required because both PCs are on the same simple local network.

Network Topology

The basic network topology used in the case study is:

PC0 —— 2960 Switch —— PC1

Case Study Procedure

Step 1 – Open Cisco Packet Tracer: Open Cisco Packet Tracer and create a blank workspace.

Step 2 – Add two PCs: Select End Devices → PC and place PC0 and PC1 in the workspace.

Step 3 – Add a switch: Select Network Devices → Switches → 2960 and place the switch between the two PCs.

Step 4 – Connect the devices: Use Copper Straight-Through cables. Connect PC0 FastEthernet0 to Switch FastEthernet0/1 and PC1 FastEthernet0 to Switch FastEthernet0/2. Wait until the link indicators become green.

Step 5 – Configure PC0: Open PC0 → Desktop → IP Configuration. Enter IP address 192.168.1.10 and subnet mask 255.255.255.0.

Step 6 – Configure PC1: Open PC1 → Desktop → IP Configuration. Enter IP address 192.168.1.20 and subnet mask 255.255.255.0.

Step 7 – Test communication: Open PC0 → Desktop → Command Prompt and type: ping 192.168.1.20. Successful replies indicate that PC0 can communicate with PC1.

Step 8 – Enter Simulation Mode: Change from Realtime to Simulation mode in Cisco Packet Tracer.

Step 9 – Generate a packet: Use Add Simple PDU, select PC0 as the source and PC1 as the destination.

Step 10 – Observe packet movement: Click Capture/Forward repeatedly and observe the packet travelling from PC0 to the switch and then to PC1.

Communication Verification

The ping operation is used to verify communication between PC0 and PC1. A successful result produces replies from 192.168.1.20. This confirms that the two devices are correctly configured and can exchange network packets.

OSI Model Analysis

Layer	Name	Function in Communication
7	Application	Provides user/application communication.
6	Presentation	Handles data format and encryption.
5	Session	Starts and manages communication sessions.
4	Transport	Provides reliable delivery using TCP/UDP.
3	Network	Uses IP addressing and routing.
2	Data Link	Uses MAC addresses and frames.
1	Physical	Transmits bits through cables and electrical signals.

Memory aid for the seven OSI layers: A P S T N D P — All People Seem To Need Data Processing.

TCP/IP Model Analysis

TCP/IP Layer	Examples / Function
Application	HTTP, DNS, FTP and application-level communication.
Transport	TCP and UDP for transport services.
Internet	IP addressing and packet delivery.
Network Access	Ethernet, Wi-Fi and physical/network access functions.

Relationship Between OSI and TCP/IP

OSI Model	TCP/IP Model
Application + Presentation + Session	Application
Transport	Transport
Network	Internet
Data Link + Physical	Network Access

Packet Analysis

In Simulation Mode, selecting the packet/envelope allows the student to inspect important information such as the source IP address, destination IP address, source MAC address, destination MAC address, protocol information, and packet movement. This demonstrates that different communication layers add or process different types of information.

Real-Life Interpretation

If PC0 sends the message "Hello" to PC1, the application creates the message. The transport layer manages the data, the network layer associates the IP addressing information, the data link layer uses MAC addresses and frames, and the physical layer carries the bits through the cable. At PC1, the process is reversed and the original message is delivered to the application.

Results

The network was successfully created using two PCs, a 2960 switch, and copper straight-through cables. PC0 and PC1 were configured with IP addresses 192.168.1.10 and 192.168.1.20 respectively. The ping test verified communication between the devices. Simulation Mode demonstrated packet movement through the network and helped relate the OSI seven-layer model to the TCP/IP four-layer model.

Conclusion

This case study provides a simple practical demonstration of network communication models. Cisco Packet Tracer makes it possible to observe packet movement without requiring physical networking equipment. The experiment shows how the OSI and TCP/IP models organize network communication into layers and how addressing, transport, and physical transmission work together to deliver data from a source computer to a destination computer.

Key Learning Outcomes

- Understand the purpose of layered network communication models.
- Configure basic IPv4 addresses in Cisco Packet Tracer.
- Use ping to test connectivity.
- Understand the seven OSI layers and their basic functions.
- Understand the four layers of the TCP/IP model.
- Observe packet movement and packet details in Simulation Mode.

Reference Material

The case study is prepared from the uploaded reference document, "Demonstration of Network Communication Models (OSI / TCP-IP) Using Cisco Packet Tracer," including its procedure, network configuration, OSI/TCP-IP tables, and packet-flow explanation.